USN | | | | | | | | | | [24MTFT101]

FIRST SEMESTER M. Tech EXAMINATION, JANUARY - 2025

ADVANCES IN FOOD TECHNOLOGY

Time: 3 Hours **Maximum Marks: 100**

Instructions:
i. Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 20 marks.
ii. Missing data may be suitably assumed.

			CO's	Blooms Level	Marks
1.	**a.**	Illustrate the categories of food industries based on its finished products.	**1**	**4**	**10**
	b.	Summarize on the operational departments in food industries and its importance.	**1**	**4**	**10**
2.	**a.**	Elaborate on the types of microorganisms found in food products and add a note on the common food pathogens.	**2**	**6**	**10**
	b.	Write a note on fermented food and microbial cultures associated with fermented foods.	**1, 2**	**6**	**10**
3.	**a.**	Appraise on the structural, physicochemical, nutritional and functional aspects of Carbohydrates and lipids.	**3**	**5**	**10**
	b.	Distinguish the food additives and its Role, classification and Numbering system.	**1, 3**	**4**	**10**
		OR			
4.	**a.**	List out the types of functional groups present in food products and add a note on its importance.	**3**	**4**	**10**
	b.	Discuss on the structural, physicochemical, nutritional and functional aspects of proteins, vitamins and minerals.	**3**	**6**	**10**
5.	**a.**	Discuss the structure of water and strengthen your answer by mentioning its importance for food products.	**4**	**6**	**10**
	b.	Distinguish the Sorption and desorption phenomenon and write a note on its isotherms.	**2, 4**	**4**	**10**
		OR			
6.	**a.**	Write down in detail about the types of water present in food products.	**2, 4**	**6**	**10**
	b.	Judge the term Water Activity in food products and write down its significance and relevance in deteriorative processes.	**2, 4**	**4**	**10**

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7.	**a.**	Contrast the principle of a pH meter and write down its importance in monitoring the food quality.	**5**	**4**	**10**
	b.	Illustrate on the working principles of a water activity meter and its significance in assessing the shelf-life of food products.	**5**	**4**	**10**
		OR			
8.	**a.**	Assess on the calorimetry to measure the energy content of food products and give an suitable example for its application in food technology.	**5**	**4**	**10**
	b.	Illustrate the molecular techniques like RT-PCR and ELISA for detecting the microbial contamination in food.	**5**	**4**	**10**